YIFEI SUN

Staff Software Engineer · Infrastructure & AI-Integrated Platform Systems · 13 Years

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SUMMARY

Staff-level infrastructure engineer with 13 years building large-scale distributed systems and integrating AI into engineering workflows. Led cross-team redesign of a high-traffic API serving millions of daily requests (p99 250ms→50ms, availability 98%→99.9%), built a federated GraphQL gateway consolidating 12 downstream services, and established architecture standards across 10+ teams and 3 departments. Built an MCP server for agent orchestration and a GenAI evaluation platform with prompt engineering, multi-provider LLM integration, and automated quality assessment. Uses AI tools daily to accelerate development. Early engineer at Toast through hypergrowth; ships fast with measurable impact.

LEADERSHIP HIGHLIGHTS

- Led cross-team redesign of highest-traffic API: p99 latency 5× reduction, 99.9% availability
- Drove architecture standards across 10+ teams, 3 departments; uses AI tools daily to optimize workflows
- Built MCP server for agent orchestration and GenAI platform with prompt engineering and LLM evaluation
- Mentored senior engineers to Principal-level scope; ships 0→1 products fast with measurable impact

Technologies: Java, Spring Boot, GraphQL (Netflix DGS), Python, Node.js · AWS, Kubernetes, Kafka, Redis, PostgreSQL · Microservices, Event-Driven, CQRS · MCP/Agent Orchestration, Prompt Engineering, LLM Evaluation · Datadog, Grafana, Distributed Tracing · CI/CD, Docker

INDEPENDENT PROJECT

GenAI Decision Engine & LLM Evaluation Platform

- Architected a hybrid rules + multi-path GenAI inference engine with prompt/context engineering, producing explainable real-time decisions with full observability, bias monitoring, and quality assessment.
- Designed a configurable policy framework with DB-backed configs and a scheduler-driven tuning loop — enabling safe, iterative prompt optimization and responsible AI deployment with measurable quality gates.
- Built an LLM evaluation system for automated accuracy/quality reviews, offline backtesting, and parameter search (grid & Bayesian optimization) across multiple LLM providers.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead Oct 2019 – Present

High-Traffic API Modernization — Drove Change Across the Organization

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions); validated at 50% traffic via controlled rollout.
- Introduced a new orchestration service and GraphQL boundary, consolidating 12 downstream integrations into 5 well-defined services, reducing fan-out and simplifying failure modes.
- Produced initial architecture in ~2 weeks; shipped the first production milestone in 6 months — ahead of schedule. Identified and resolved performance and scalability bottlenecks throughout.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Built a parity-testing framework for automated quality and accuracy validation; defined architectural standards and reduced years of technical debt with clear documentation.

GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to avoid JIT compilation latency spikes on cold starts.

MCP Server — Agent Orchestration & Tool Calling (0→1)

- Sole backend engineer designing and implementing an MCP server for agent orchestration, exposing CarGurus capabilities as tools callable by AI agents (ChatGPT via OpenAI Apps SDK).

- Applied prompt/context engineering to optimize agent-tool interactions; designed for safe, incremental rollout with quality monitoring and responsible AI practices.

Digital Shopping & Online Leads Products

- Led backend design for a new digital shopping product launched during the pandemic, using microservice and event-driven architecture for inventory, merchandising, and lead flows.
- Subsequently led backend work for additional commerce experiments, enabling rapid product launches without destabilizing core systems.

Cross-Org Platform & API Leadership

- Established patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and decision records driving consensus across platform and product teams.
- Mentored multiple senior engineers on system design, performance work, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future growth.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010